



Potential use of *Tinospora cordifolia* as a herbal supplement in dairy animals: a review

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Abstract

Supplementation is an integral part of modern dairy-feeding practices which mainly emphasise on higher production. Different supplements have been utilized by the farmers according to the need and action of supplements. *Tinospora cordifolia* (TC) is one such herbal supplement which can be utilized to feed dairy animals either alone or in combination with other supplements. TC is a herb found in tropical countries and having number of medicinal properties. Beneficial health effects of TC include immunostimulation, protection against inflammation and bacterial action, hepatoprotection, antioxidant and antineoplastic effects. Available studies on TC supplementation in dairy animals have shown a positive effect on health, body parameters and production performance. Supplementing TC is economically feasible, and it can be adopted by farmers as it helps in improving their net income. However, more research is needed in the field of dairy animals to explore the full potential of this herb. This review is written with the objective of highlighting the possibilities of using TC in dairy animal rations and promoting research in the concerned field to fill the gaps in knowledge. Thorough knowledge about TC and its effect in dairy animals can add a new concept to the supplementation process and developments of dairy rations.

Keywords Dairy cattle · Herb · Immunostimulant · Ruminants · Supplementation · *Tinospora*

Introduction

Dairy farming is one of the advancing sectors of agriculture with main aim of profitability. Dairying is essential for provision of employment to majority of resource poor population in developing countries and also for economy of the nation. Recent advances are happening in dairy farming rapidly aiming more production with locally available resources. Herbal supplements are one of these recent advances, and nowadays, these are more focused because of several advantages (Sriranga et al. 2021). Although a number of herbal products have been traditionally employed in dairy animal rations, these were less exploited on commercial scale.

Presently with the more health concerns and more awareness among consumers, herbal supplements are more emphasised as these are natural and having least side effects. Herbal supplements with well-established therapeutic and prophylactic effects can be used as safe and effective alternative to allopathic medicine (Rehman et al. 2016). *Tinospora cordifolia* (TC) is one such herbal supplement having multiple advantages which can be utilized by the farmers to feed dairy animals.

TC which is also famous as Guduchi, Giloy, Amrita, Gurach and *Tinospora* belongs to family *Menispermaceae* and is a tropical herb that can rejuvenate. The full taxonomical classification of this herb is depicted in Table 1. This herb has been described as “*Rasayana*” and has attained prime position in “*Ayurvedic Materia Medica*” (Bhattacharyya and Bhattacharyya 2013). The use of this herb as tonic (Sinha et al. 2004), immunomodulatory effects (Aranha et al. 2012), antioxidant properties (Sengupta et al. 2011), antibacterial uses (Mukherjee et al. 2010), hepatoprotective effect (Sharma and Pandey 2010) and anti-inflammatory properties (Krishna et al. 2009) has been well known in ayurveda and medicine. The employment of TC as fodder for livestock

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Table 1 Taxonomical classification of *Tinospora cordifolia*

Kingdom	Plantae
Division	Magnoliophyta
Class	Magnoliopsida
Order	Ranunculales
Family	Menispermaceae
Genus	<i>Tinospora</i>
Species	<i>T. cordifolia</i>

is widely accepted as it is found rich in nutrients like carbohydrate, protein and minerals (Nile and Khobragade 2009). Proximate analysis of TC revealed that it contains 66.40% dry matter and 92.15% of organic matter. Proportions of different components in TC is calculated as 7.50% crude protein, 6.89% ether extract, 57.10% crude fibre, 20.66% nitrogen free extract and 7.85% ash (Raj et al. 2020). The higher amount of important minerals like phosphorus (P), potassium (K), sodium (Na), calcium (Ca), iron (Fe), zinc (Zn) and magnesium (Mg) in TC adds to its nutritive value and increases the utility of herb (Vijayakumari and Thirumurugan 2018).

TC is having multidimensional properties which aids to health of animal. Out of the whole plant, it is the stem that has higher medicinal properties (Mittal et al. 2014). Glycosides, alkaloid and steroids are the major components of TC that have anticomplementary and immunostimulating activity (Kapil and Sharma 1997). The activity for immunomodulation has been tested and reported in the alcoholic and aqueous extract of TC in different animal models (Manjrekar et al. 2000; Aher and Wahi 2012). There are a number of active compounds that have been isolated from different parts of this plant including alkaloids, diterpenoid lactones, glycosides, steroids, sesquiterpenoid, phenolics, aliphatic compounds and polysaccharides (Singh et al. 2003). All of these play a role in prevention and curing of diseased conditions by different mechanism of actions (Saha and Ghosh 2012). TC has a cure for fever due to its diverse medicinal properties (Srivastava 2011); it has also been used to treat other ailments like weakness, dysentery, dyspepsia infections (Singh et al. 2003), secondary syphilis, impotency, viral hepatitis (Neeraja and Margaret 2013), gout (Bhattacharyya and Bhattacharyya 2013), anaemia rheumatoid arthritis, jaundice and diabetes (Chaudhari and Shaikh 2013; Tamboli et al. 2013). The herb is available in powder or dried form due to its pharmacological properties and is therefore given in diet of animals for prevention of diseases (Yanshi and Rahal 2019). Also, plants containing phenolic compounds have a good influence on rumen fermentation and increase ruminant output efficiency (Unnawong et al. 2021; Sommai et al. 2021).

Although TC is well studied for its properties with different animal models (Goel et al. 2014; Rakib et al. 2020), its potential is not explored fully in the field of dairy animals. Dairy animals differ in physiology and metabolism due to the

presence of rumen (Clauss and Hummel 2017). TC is having different active compounds, and their effect of the ruminal metabolism still needs to be explored. Furthermore, research is required regarding the quantity of TC that is needed to be fed, i.e., dose rate needs to be standardized in dairy animals. Also, the effect of individual active component of TC needs to be studied along with their mechanism of actions in order to utilize the full potential of TC. After extracting complete information regarding TC, it can be utilized in the development of new supplementary products for dairy animals. In the upcoming sections, all the information regarding properties of TC and its effect on different parameters related to production in dairy animals were described, which can develop a new insight into development of dairy rations.

Health effects of *Tinospora cordifolia*

Immunomodulatory effect

Production of cytokines, mitogenicity, stimulation and activation of immune effector cells is highly influenced by aqueous extract of TC, and these were reported well in B16F10 mouse melanoma cells (Upadhyaya et al. 2011). Purandare and Supre (2007) reported a significant improvement in wound healing when TC was employed as an adjuvant in surgical treatment of diabetic foot ulcer patients. Reports of increased phagocytosis in vitro against sheep red blood cells and antibody production in rats were found with aqueous and ethanolic extracts of TC stem powder (Ranjith et al. 2008). TC also enhances the concentration of IgG antibodies. Antibody production was directly related to the dose of active principles of TC, i.e., antibody production was significantly higher at higher concentrations of TC (Kapil and Sharma 1997).

A polysaccharide α -D-glucan found in TC, at concentration of 100 μ g/ml, leads to activation of different subsets of the lymphocytes such as NK (natural killer) cells (331%), T cells (102%) and B cells (39%). Immune activation in normal lymphocytes by α -D-glucan at 100 μ g/ml concentration evoked the synthesis of different interleukins (IL-1 β , IL-6, IL-12 p70, IL-12 p40, IL-18, IFN- γ), TNF- α (tumour necrosis factor) and MCP-1 (monocyte chemoattractant protein). Complements in the alternate pathway were also activated by this (Nair et al. 2004).

Anti-inflammatory and antibacterial effect

Reports of enhanced tolerability and the absence of serious adverse reactions in allergic rhinitis patients by the action of aqueous extract of TC were proven by significant reduction in symptoms of allergic rhinitis which includes sneezing, nasal discharge, nasal obstruction and nasal pruritus (Badar et al. 2005). The effectiveness of TC in acute inflammation was

found higher than the acetyl salicylic acid when given in rats having hind paw oedema and cotton pellet granuloma (Utpalendu et al. 1999). The potential of methanol extracts of TC was also explored against microbial infections and positively found to be inhibiting the microbial growth (Narayanan et al. 2011). Proof of enhanced anti-inflammatory and antibacterial effect was also seen from the reports of increased phagocytic activity of neutrophils, reduced somatic cell count, reduced total bacterial count and enhanced lysosomal enzyme content of the milk polymorphonuclear cells in bovine subclinical mastitis on intramammary infusion of hydro-methanolic extracts of TC (Mukherjee et al. 2010). Leaves of TC were also tested and observed against a test microorganism *Escherichia coli* using three different solvent extracts, i.e. ethanolic, methanolic and aqueous extracts of the leaf, and these were reported to have a good zone of inhibition against *E. coli* in slip disc method (Kumar et al. 2017).

Hepatoprotective effect

The hepatoprotective effect of TC was shown by the enhanced Kuffer cell activity in chronic liver disease model study in rats (Nagarkatti et al. 1994). Adhvaryu et al. (2007) also proved hepatoprotective results of TC in guinea pigs by conducting a trial in which hepatic injury is induced to the guinea pigs and then treated with TC and found that TC is responsible for maintaining normal liver histology and liver enzyme levels. A report by Bishayi et al. (2002) stated that mature albino rats intoxicated with carbon tetrachloride was given hepatoprotection by TC extract, and this was proved by the normal values of alanine transaminase (ALT), aspartate aminotransferase (AST), serum alkaline phosphatase and serum bilirubin obtained after treating with TC. The extract of TC has also been shown to display property to inactivate hepatitis B and E surface antigen in in vitro technique in 48–72 h (Mehrotra et al. 2002).

Antioxidant effect

The antioxidant activity of TC is attributed to its alkaloids, flavonoid and phenol contents. Due to this (antioxidant) property of TC, an enhanced activity of erythrocytes membrane lipid peroxide and catalase was observed in diabetic rats and also the activity of superoxide dismutase and glutathione peroxidase enzymes got reduced (Sivakumar and Rajan 2010). Antioxidant activity of TC also helps in lessening the chemotoxicity caused by free radicals (Rawal et al. 2004). TC has also been reported to scavenge the free radicals produced during aflatoxicosis and provides safety against aflatoxin-induced nephrotoxicity in mice (Gupta and Sharma 2011).

Antineoplastic effect

Most of the antineoplastic responses of TC were analyzed in animal models. Jagetia and Rao (2006) showed a response of TC extract similar to that of doxorubicin in Hela-cultured cells and stated that it is having dose-dependent cytotoxic effect. TC extract was also found to diminish tumour mass and enhanced survival time on administration in mice with induced carcinomas (Leyon and Kuttan 2004). Increase in proliferation of bone marrow cells in both mice having tumour and normal mice was also reported on administration of alcoholic extract of TC (Singh et al. 2006).

Other beneficial effects

Studies on TC also showed its antidiabetic effects in rats confirmed by reduction in their blood glucose levels (Nagaraja et al. 2008; Rajalakshmi et al. 2009). Methanolic extract of TC also found to hasten the wound healing in albino mice (Nema et al. 2012). TC also reported to possess analgesic activity which was confirmed by the experiment on albino rats (Goel et al. 2014). As per experiment, analgesic activity probably involves peripheral (inhibition of prostaglandin synthesis) as well as central mechanisms (opioid receptors).

Use of *Tinospora cordifolia* in dairy animals

In dairy animals, TC is fed at different dose rates by researchers in different species and various positive results were reported in the performance and health status of animals. It was used as a supplement, prophylactic medicine and also for curing diseases in dairy animals by different workers. It can be fed to dairy animals either alone or can be mixed with other supplements depending on its interaction with them. Different proposed effects of TC in dairy animals include improved immunity (Gupta et al. 2016; Rajneesh et al. 2021), increased production (Mallick and Prakash 2012; Mir et al. 2015), improved body condition (Patel et al. 2017) and reduced stress (Rajneesh et al. 2021), and these are depicted in Fig. 1. Although the whole plant of the TC got different chemical constituents and active principles, most of the chemical constituents are concentrated in the stems and leaves (Garg and Garg 2018), and hence, these are utilized for feeding animals. Different chemical constituents along with their location in TC herb are presented in Table 2.

Mir et al. (2014) supplemented TC stem powder to Murrah buffaloes at a dose rate of 120 g/day/buffalo and found significant ($P < 0.05$) increment in milk yield and enhanced milk quality, and the possible reason quoted for this is the regulation of stress due to heat, increase in milk secretion, reduced infection in mammary gland, improved utilization of nutrients and increased immune level of buffaloes by

Fig. 1 Pictorial representation of utility and effects of *Tinospora cordifolia* in dairy animals

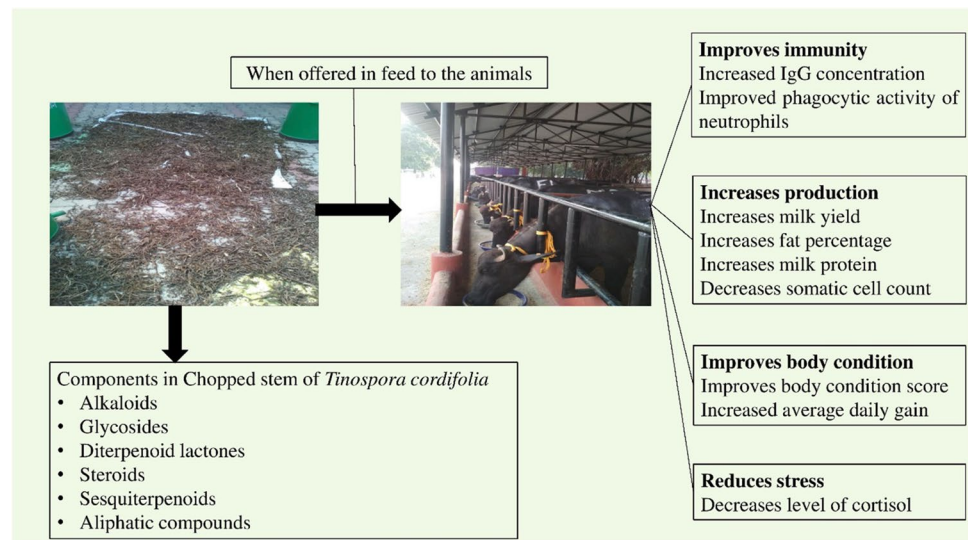


Table 2 Chemical constituents of *Tinospora cordifolia* and their distribution in different parts of plant

Type of chemical	Active principle	Distribution in plant	References
Alkaloids	Berberine	Stem	Upadhyay et al. (2010), Sharma et al. (2012)
	Tinosporin	Leaf	
	Tinosporic acid	Whole plant	
	Isocolumbin	Root	
	Palmitine	Stem, root	
Glycosides	Cordifoliside, Tinocordiside, Tinocordifoliside, Cordioside, Syringin, Syringinapiosylglycoside, Palmatosides, 18 Nonderodane glycoside	Stem	Kapil and Sharma (1997), Sinha et al. (2004), Bala et al. (2015)
Diterpenoid lactones	Tinosporon, Furanolactone, Clerodane derivatives, Jateorine, Tinosporisides	Whole plant	Choudhary et al. (2013), Wink (2015)
	Diterpenoid, Columbin	Stem	
Steroids	Sitosterol	Aerial part, stem	Singh and Chaudhuri (2017), Sharma et al. (2019)
	Octacosanol, Heptacosanol, Ecdysterone, Giloinsterol	Stem	
Sesquiterpenoids	Tinocordifolin	Stem	Maurya and Handa (1998)
Other miscellaneous compounds	Jatrorrhizine	Root	Singh et al. (2003), Kumar et al. (2017)
	Tinosporidin, Cordifol, Cordifellone, Giloin, Giloinin	Whole plant	

TC supplementation. Gupta et al. (2016) proved the immunostimulant activity of TC in dairy cattle by supplementing dried and grounded TC stem powder at a dosage of 100 mg/kg body weight mixed in concentrate fed to the animal for a period of 5 days. They reported significant increase in total serum immunoglobulin (IgG) and mean phagocytic index values in the blood profile of animals.

Mallick and Prakash (2012) conducted an experiment on dietary supplementation of guduchi (TC) in crossbred Karan Fries cattle, and feeding of cows was done at the rate of 60 g/day and 120 g/day for 45 days before and after parturition, respectively. The results of the experiment showed a positive

response on production and reproduction performance of supplemented crossbred cows with an increase of 1445.4 kg lactation yield and decrease in incidence of reproductive disorders in comparison to the control group. Supplementation of Jersey crossbred cows at the rate of 150 g/cow/day was also reported, and it was reported to significantly affect the blood glucose levels of cows as lower levels of blood glucose were found in TC-fed cows in comparison to control group cows (Sharma et al. 2018).

A higher dose rate of TC was offered in goats in comparison to large ruminants, and positive results were observed. Goats were fed with TC powder at the rate of 500 mg/kg body weight

daily for 30 days, and the antistress effect of TC on solar radiation-induced heat stress was studied. It was found that TC-supplemented animals have shown significant ($P < 0.05$) reduction in heat stress, which was depicted by change in the antioxidant profile. There is modulation in peroxidation of lipid by provision of TC plant powder with decrease in level of glutathione and superoxide dismutase to normal in goats that are in heat stress (Devasena and Adilaxmamma 2014). TC was also supplemented in Gaddi goats at the rate of 5% dry matter intake (DMI) to study the role of TC in ameliorating heat stress, and the results of the study confirm that TC supplementation helps in eradicating heat stress to some extent (Soni et al. 2018).

Effect of *Tinospora cordifolia* supplementation on body weight of dairy animals

TC is reported to improve the body weight gain and growth in young and growing dairy animals, i.e. calves, whereas in adult animals (cows and buffaloes), some reports suggest that there was no significant effect whereas others suggest that there is decrease in rate of reduction in body weight during lactation. Improvement in weight gain by the supplementation of TC is attributed to its stomachic action (Sinha et al. 2004) and because of that, there is improved dry matter intake (Mir et al. 2014). Patel et al. (2017) conducted a trial on 14-day-old Murrah buffalo calves by supplementing them with a self-made mixture of *Emblia officinalis* and TC in the ratio of 2:1 at a dose of 450 mg/kg body weight once per day orally for 28 days. They found a significant difference ($P < 0.05$) in body weight gain of supplemented group (8.5 kg) calves over non-supplemented group calves (5.5 kg). Another study on cross-breed calves showed that self-made rumenotonic drug which is having TC as one of the constituents (*Apium graveolens*, *Amomum subulatum*, TC, *Cuminum cyminum*, *Piper nigrum*, *Piper longum*, *Plumbago rosea*, *Pmbelia ribes*, yeast, ammonium chloride and black salt) when fed to the calves improves the appetite and digestion process and also stimulates the growth of calves (Gautam et al. 2013). In adult Murrah buffaloes when supplementation of TC was done at a dose rate of 150 g per day, there was no significant effect observed in the body weight of buffaloes (Rajneesh et al. 2020a).

Effect of *Tinospora cordifolia* supplementation on dry matter intake of dairy animals

Dry matter of dairy animals is positively influenced by the TC supplementation. Mir et al. (2014) reported an increase of 4.77% in dry matter intake (DMI) of TC-supplemented group in comparison to control group Murrah buffaloes. Reports from Patel et al. (2013) also showed significantly

($P < 0.05$) higher dry matter intake per 100 kg body weight (DMI%) in buffaloes on supplementation of polyherbal galactagogue containing TC. Another investigation of TC supplementation at a dose rate of 1 g/kg body weight in Muzzafarnagari rams showed a significant ($P < 0.05$) increase in mean daily intake of supplemented rams (Jayaganthan et al. 2013). Similar report of improved feed intake was reported in Bengal goats on supplementation of TC (Sahoo et al. 2001).

Effect of *Tinospora cordifolia* supplementation on milk production and milk composition of dairy animals

TC is a well-known herbal galactagogue and traditionally used by Indian farmers to improve milk production in dairy animals (Dandotiya et al. 2013). Different researchers studied the effect of TC supplementation in dairy animals, and majority of the works showed that it helps in increasing the milk production. Reports of increased milk yield in dairy animals on supplementation of TC are described in Table 3. Milk composition is not affected significantly by the TC supplementation; however, there are few reports of significant change in milk components. Reports of Mallick and Prakash (2012) showed significant increase ($P < 0.05$) in milk yield of Karan Fries crossbred cows over 305 days of lactation by supplementing ration of cows with TC but no significant difference was reported in milk composition (fat, protein, lactose and SNF). Similar reports of significant improvement in the milk yield of Murrah buffaloes were reported on TC supplementation (Mir et al. 2014; Rajneesh et al. 2020b). In milk composition, Rajneesh et al. (2020b) showed a significant improvement in milk fat, i.e. 0.49% of more milk fat in comparison to control group (7.85 vs 7.36%), whereas Mir et al. (2015) reported an improvement in the milk protein of the supplemented buffaloes in comparison to control group (3.8 vs 3.74). In contrary to this, Sharma et al. (2017) observed no significant effect on milk production and milk composition on supplementation of TC in Jersey crossbred cows.

Effect of *Tinospora cordifolia* supplementation on somatic cell count of dairy animals

An important indicator of udder infection is somatic cell count (SCC) in milk (de Haas et al. 2002). Increase in the SCC above the normal values ($> 200,000$ cells/ml) is indicative of ongoing infection or any physiological problem (Bradley and Green 2005). Suggestion from Aranha et al. (2012) projects the

Table 3 Effect of *Tinospora cordifolia* supplementation on milk production of dairy animals

Animal species	Details	TC (dose per animal)	Milk yield (kg)		References
			Control group	Supple-mented group	
Murrah buffalo	Average daily milk yield*	150 g/day	805	10.30	Rajneesh et al. (2020b)
Jersey crossbred cows	Average daily milk yield	150 g/day	6.33	6.65	Sharma et al. (2017)
Murrah buffalo	Average daily milk yield*	120 g/day	7.19	8.00	Mir et al. (2015)
Karan Fries crossbred cows	Lactation yield*	60 g/day for 45 days before parturition and 120 g/day for 45 days after parturition	3735.4	5180.8	Mallick and Prakash (2012)

*Differs significantly between control and supplemented groups at ($P < 0.05$) level of significance

immunomodulatory activity of TC that triggers the activation of macrophages which is the result of an immunostimulatory protein (ImP) present in the stem of the plant and is proved to significantly reduce the SCC in the milk of animal supplemented with TC. Intramammary infusion of hydro-methanolic extract of TC in a study by Mukherjee et al. (2010) showed that SCC in cattle having subclinical mastitis initially raised but after 15 days of treatment, there was a significant reduction in cell count. Ameliorative potential of TC was explored by Rajan et al. (2007) through stem extracts in bovine clinical mastitis, and results revealed higher recovery rate in cows which were supplemented with extract of TC in comparison with those treated with antibiotics alone. Mukherjee et al. (2010) demonstration on therapeutic and immunomodulatory potential of stem extract of TC in bovine mastitis showed a decrease in SCC and total bacterial count. The level of SCC ($\times 10^6/\text{ml}$) in milk of animals supplemented with TC (1.37 ± 0.10) was significantly ($P \leq 0.05$) lower in comparison to the control group (1.52 ± 0.07) of Murrah buffaloes (Mir et al. 2014). A marked decrease in SCC was noted by Mallick and Prakash (2012) when cows were fed with TC throughout the trial period.

Effect of *Tinospora cordifolia* supplementation on blood parameters of dairy animals

Blood parameters denote the ongoing changes inside animals' body, and analysis of blood parameters after supplementation of TC carried out by different workers proved the beneficial effect of TC on animals' health. Rajneesh et al. (2020c) evaluated the blood profiles of lactating Murrah buffaloes after supplementing with TC and found that a significant increase was there in the values of total leukocyte count (TLC), neutrophil count, neutrophil/lymphocyte ratio (N/L) and platelet counts whereas values of red blood cells

(RBCs), haemoglobin (Hb), packed cell volume (PCV), lymphocytes and mix cells were not changed significantly by supplementation. Significant change in the values of TLC, neutrophil count, N/L and platelet counts within the normal limits is indicative of ongoing immunity responses. Similar reports of higher lymphocyte count, total leukocyte count, neutrophil count and neutrophil/lymphocyte ratio in the guduchi (TC) supplemented treatment group in comparison to non-supplemented crossbred Karan Fries cows throughout the experimental period (Mallick and Prakash 2012). Supplementation of TC in crossbred growing calves is reported to affect differently from the adult animals as it leads to increase haemoglobin, packed cell volume and neutrophil count (Ritu 2018).

Supplementation of TC also affects the concentration of other blood metabolites and hormones which indirectly affects the health of animal. Blood glucose levels of Murrah buffaloes were found significantly higher (50.91 mg/dl) on supplementation of TC in comparison to non-supplemented buffaloes (46.64 mg/dl) (Mir et al. 2014). In the same group of Murrah buffaloes, non-esterified fatty acid (NEFA) concentration was also analyzed, and researchers found that supplemented buffaloes were having significantly lower values to those of control group buffaloes (213.1 $\mu\text{M}/\text{ml}$ vs 274.3 $\mu\text{M}/\text{ml}$) indicating better energy balance on supplementation (Mir et al. 2014). Supplementation also helps in eradicating stress in animals, and this was also shown by Mir et al. (2014) who determine the levels of cortisol hormone and found significantly lower values in supplemented group (5.70 ng/ml) as compared to non-supplemented group buffaloes (6.18 ng/ml). In the same study, somatostatin hormone was also analyzed and found to be significantly higher in the supplemented buffaloes (12.12 ng/ml vs 9.06 ng/ml). Lower cortisol levels explained the lower stress in animals after TC supplementation whereas higher somatostatin levels were in response to increased blood glucose levels. Effect

on TC supplementation on blood metabolites of crossbred jersey cows was also studied, and no significant change was observed in total protein and renal profile (urea nitrogen and creatinine) of treated cows. However, their blood glucose levels have significantly reduced in comparison to control group cows which is contrary to the results of Mir et al. (2014), and researchers have attributed this effect to the increased demand of glucose by mammary tissues on increased milk production (Sharma et al. 2018). Blood glucose levels depend on several factors like type and amount of feed given to animals and stage of lactation. When more glucose is available to mammary tissue, it can be converted to lactose, and hence, milk production increases. TC is a potent galactagogue, and glucose-sparing action of TC is one of the reasons for this. However, more studies are required to clearly mention the effect of TC supplementation on blood glucose levels of dairy animals.

Effect of *Tinospora cordifolia* on immunity and health of dairy animals

TC is a potent immunostimulant, and it has certain positive influence on immune status and health of dairy animals. The role of TC in immunomodulation was observed when local non-descript cow was supplemented with this herb at the rate of 100 mg/kg body weight in concentrate mixture for a period of 5 days (Gupta et al. 2016). Increased phagocytic activity of milk polymorphonuclear cells was reported by Mukherjee et al. (2010) in cows that were in diseased condition and were provided with TC extract as a treatment. Supplementation of TC in Murrah buffaloes also leads to a significant increment in the immunoglobulin levels (IgG) in comparison to non-supplemented buffaloes indicating better immunity of supplemented buffaloes (Rajneesh et al. 2021).

Economics of supplementing *Tinospora cordifolia* in dairy animals

Any supplementation is only effective and adopted by farmers when it is economically feasible. Calculation of cost benefit ratio on supplementation of TC in Murrah buffaloes was done by Rajneesh et al. (2020b), and they have found a ratio of 2.79 on complete budgeting of the trial. The increase in profits was from increased milk production and decreased service period of the TC-supplemented buffaloes. A value of cost benefit ratio above one marks the economic feasibility of supplementation. Similarly, higher net profit was shown on supplementation of TC in comparison to control by Mallick (2010).

Conclusion

TC possesses multidimensional properties because of the presence of different active principles which can be utilized in dairy animals. TC supplementation practices in dairy animals showed positive effect on health of animals by improving immunity of the animals and also improves the body condition and growth rate. It is also effective in enhancing milk yield and quality of milk by improving energy balance and eradicating stress in dairy animals. With the available information, it is clear that TC is having a huge potential to be used as a supplement in dairy industry, but still some of the gaps need to be bridged through extensive research. As different workers supplemented TC at varying dose rates in different animals and reported significant improvements, more research is needed to standardize the dose rate. Further extensive research on clinical usage with long-term feeding trials is required for recommendation of TC as a herbal supplement with readily available ready-made preparations.

Author contribution All the authors work in coordinated way in gathering the information on the topic and help in preparing the manuscript.

Data availability Information is collected from different research papers and other sources; they were cited in the reference, and they can also be provided by the author on request.

Declarations

Ethics approval As this article is a review paper, no ethical approval is taken. Although, all the data used were properly cited in the references.

Consent to participate All the authors were aware regarding participation and publications.

Consent for publication All the authors were aware regarding participation and publications.

Conflict of interest The authors declare no competing interests.

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